

Roll no :- S002	Name :- Kaif Ansari
Batch :- 1	Class :- SY.BSC.IT
Date of assignment :- 18/08/2026	Date of submission:-18/08/2026

PRACTICAL 19

Aim:- Combine two plots (e.g., scatter and line plot) in a single visualization using Breeze-viz.

Input:-

```
import breeze.linalg._
import breeze.plot._

object Kaif1 extends App {
  val x = DenseVector(1.0, 2.0, 3.0, 4.0, 5.0) // Sample X-axis values
  val yscat = DenseVector(2.0, 4.0, 1.0, 8.0, 7.0) // Y values for scatter
  val yline = DenseVector(1.5, 3.0, 4.5, 6.0, 7.5) // Y values for line plot

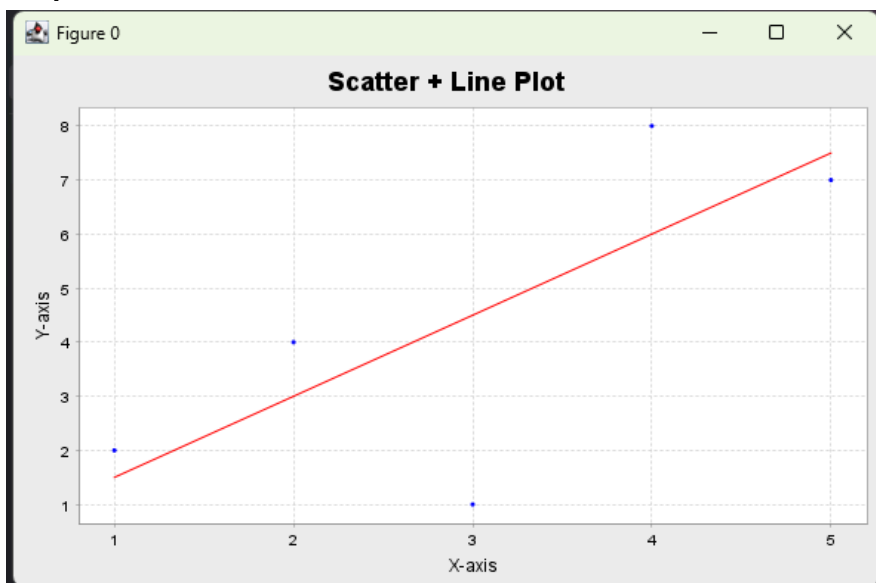
  val fig = Figure() // Create a new figure
  val plt = fig.subplot(0)

  plt += plot(x, yscat, style = '.', colorcode = "blue")
  plt += plot(x, yline, style = '-', colorcode = "red")

  plt.title = "Scatter + Line Plot"
  plt.xlabel = "X-axis"
  plt.ylabel = "Y-axis"

  fig.refresh()
}
```

Output:-



PRACTICAL 20

Aim:- Read a CSV file and calculate basic statistics for each numeric column. Use the scala-csv library or similar tools.

Input:-

```
import scala.io.Source

object Kaif1 extends App {
  val file = Source.fromResource("iris.csv")
  val data = file.getLines()
    .drop(1)
    .map(line => line.split(",")(0).toDouble)
    .toList

  file.close()

  val count = data.length
  val mean = data.sum / count
  val min = data.min
  val max = data.max

  val variance = data.map(x => math.pow(x - mean, 2)).sum / count
  val std = math.sqrt(variance)

  println(s"Count: $count")
  println(f"Mean: $mean%.2f")
  println(f"Minimum: $min%.2f")
  println(f"Maximum: $max%.2f")
  println(f"Standard Deviation: $std%.2f")
}
```

Output:-

```
Count: 150
Mean: 5.84
Minimum: 4.30
Maximum: 7.90
Standard Deviation: 0.83

Process finished with exit code 0
```